CONVERGENCE OF DECENTRALIZED GRADIENT DESCENT WITH RANDOMIZED QUANTIZATION

1. Modified Algorithm with Randomized Quantization

We consider a network of n agents collaborating to solve the consensus optimization problem $\min_{x\in\mathbb{R}^p} \sum_{i=1}^n f_i(x)$. The decentralized gradient descent (DGD) algorithm is modified to account for communication constraints, where agents exchange quantized versions of their local state vectors.

Definition 1 (Random Quantizer)**.** A random quantizer $Q(x)$ maps a vector $x \in \mathbb{R}^p$ to a discrete set of representation symbols. Let each component x_d of x be in an interval $[\ell_d, u_d]$ partitioned into bins of width Δ_d. The quantizer $Q(x)$ acts component-wise and has the following properties, conditioned on x:

$$\mathbb{E}[Q(x)|x] = x \quad \text{(Unbiased)} \tag{1}$$

$$\mathbb{E}[\|Q(x) - x\|^2|x] \leq \frac{p\Delta^2}{4}, \quad \text{where } \Delta = \max_d \Delta_d \quad \text{(Bounded Variance)} \tag{2}$$

We define the quantization error as $\varepsilon(x) = Q(x) - x$.

The DGD update rule at each agent i is modified as follows:

$$x_{(i)}(k+1) = \sum_{j=1}^n w_{ij} Q(x_{(j)}(k)) - \alpha \nabla f_i(x_{(i)}(k)) \tag{3}$$

where $W = [w_{ij}]$ is a symmetric, doubly stochastic mixing matrix.

Letting $\varepsilon_{(j)}(k) = Q(x_{(j)}(k)) - x_{(j)}(k)$, we can rewrite the update as:

$$x_{(i)}(k+1) = \left(\sum_{j=1}^n w_{ij} x_{(j)}(k) - \alpha \nabla f_i(x_{(i)}(k)) \right) + \sum_{j=1}^n w_{ij} \varepsilon_{(j)}(k) \tag{4}$$

Let $x_{\text{vec}}(k) \in \mathbb{R}^{np}$ be the stacked vector of all local states $[x_{(1)}(k); \ldots; x_{(n)}(k)]$. Similarly, let $h(k) \in \mathbb{R}^{np}$ be the stacked gradients and $\varepsilon_{\text{vec}}(k) \in \mathbb{R}^{np}$ be the stacked quantization errors. The global update is:

$$x_{\text{vec}}(k+1) = (W \otimes I)(x_{\text{vec}}(k) + \varepsilon_{\text{vec}}(k)) - \alpha h(k) \tag{5}$$

Our analysis will be conducted in expectation, conditioned on the filtration $\mathcal{F}_k = \sigma(x_{\text{vec}}(0), \ldots, x_{\text{vec}}(k))$.

2. Main Proofs

2.1. Bounded Expected Gradients.

Our first key result is to show that the expected squared norm of the gradients remains uniformly bounded over time. This ensures the stability of the stochastic algorithm.

Theorem 1 (Bounded Expected Gradients)**.** *Assume each function f_i is strongly convex with modulus $\mu_f > 0$ and has a Lipschitz continuous gradient with constant $L_{f_i} \leq L_h$. If the stepsize α satisfies $\alpha < (1 + \lambda_n(W))/L_h$, then there exists a constant D_q^2 such that for all $k \geq 0$:*

$$\mathbb{E}[\|h(k)\|^2] \leq D_q^2$$

Proof. We use a Lyapunov stability argument. Define the Lyapunov function $\xi_\alpha : \mathbb{R}^{np} \to \mathbb{R}$ as:

$$\xi_\alpha(x_{\text{vec}}) := \frac{1}{2} x_{\text{vec}}^T (I - W \otimes I) x_{\text{vec}} + \alpha \sum_{i=1}^{n} f_i(x_{(i)})$$

The gradient of ξ_α is $\nabla \xi_\alpha(x_{\text{vec}}) = (I - W \otimes I) x_{\text{vec}} + \alpha h(k)$. The gradient $\nabla \xi_\alpha$ is Lipschitz continuous with constant $L_{\xi_\alpha} = (1 - \lambda_n(W)) + \alpha L_h$. The stepsize condition ensures $L_{\xi_\alpha} < 2$. The function ξ_α is also strongly convex with modulus $\mu_\xi > 0$.

The update step $x_{\text{vec}}(k+1) - x_{\text{vec}}(k)$ can be written as:

$$\begin{aligned} x_{\text{vec}}(k+1) - x_{\text{vec}}(k) &= (W \otimes I - I) x_{\text{vec}}(k) - \alpha h(k) + (W \otimes I) \varepsilon_{\text{vec}}(k) \\ &= -\nabla \xi_\alpha(x_{\text{vec}}(k)) + \delta(k) \end{aligned}$$

where $\delta(k) = (W \otimes I) \varepsilon_{\text{vec}}(k)$. Note that $\mathbb{E}[\delta(k) | \mathcal{F}_k] = 0$.

Using the descent lemma for ξ_α and taking conditional expectation:

$$\mathbb{E}[\xi_\alpha(x_{\text{vec}}(k+1)) | \mathcal{F}_k] \leq \xi_\alpha(k) + \mathbb{E}[\langle \nabla \xi_\alpha(k), x_{\text{vec}}(k+1) - x_{\text{vec}}(k) \rangle | \mathcal{F}_k] + \frac{L_{\xi_\alpha}}{2} \mathbb{E}[\| x_{\text{vec}}(k+1) - x_{\text{vec}}(k) \|^2 | \mathcal{F}_k]$$

The inner product term simplifies to $-\|\nabla \xi_\alpha(k)\|^2$. The squared norm term becomes $\|\nabla \xi_\alpha(k)\|^2 + \mathbb{E}[\|\delta(k)\|^2 | \mathcal{F}_k]$. This yields:

$$\mathbb{E}[\xi_\alpha(k+1) | \mathcal{F}_k] \leq \xi_\alpha(k) - \left(1 - \frac{L_{\xi_\alpha}}{2}\right) \|\nabla \xi_\alpha(k)\|^2 + \frac{L_{\xi_\alpha}}{2} \mathbb{E}[\|\delta(k)\|^2 | \mathcal{F}_k]$$

The quantization noise term is bounded as:

$$\mathbb{E}[\|\delta(k)\|^2 | \mathcal{F}_k] = \mathbb{E}[\|(W \otimes I) \varepsilon_{\text{vec}}(k)\|^2 | \mathcal{F}_k] \leq \|W\|^2 \mathbb{E}[\|\varepsilon_{\text{vec}}(k)\|^2 | \mathcal{F}_k] \leq 1 \cdot \frac{np\Delta^2}{4}$$

Let $c_d = 1 - L_{\xi_\alpha}/2 > 0$ and $C_q = (L_{\xi_\alpha}/2)(np\Delta^2/4)$. Taking the full expectation:

$$\mathbb{E}[\xi_\alpha(k+1)] \leq \mathbb{E}[\xi_\alpha(k)] - c_d \mathbb{E}[\|\nabla \xi_\alpha(k)\|^2] + C_q$$

By strong convexity of ξ_α, we have $\mathbb{E}[\|\nabla \xi_\alpha(k)\|^2] \geq 2\mu_\xi (\mathbb{E}[\xi_\alpha(k)] - \xi_\alpha^*)$, where ξ_α^* is the minimum of ξ_α.

$$\mathbb{E}[\xi_\alpha(k+1)] \leq (1 - 2c_d \mu_\xi) \mathbb{E}[\xi_\alpha(k)] + 2c_d \mu_\xi \xi_\alpha^* + C_q$$

This is a linear recurrence for $V_k = \mathbb{E}[\xi_\alpha(k)]$. Since $1 - 2c_d\mu_\xi < 1$, V_k is uniformly bounded for all k. Let this bound be B_ξ, so $\mathbb{E}[\xi_\alpha(k)] \leq B_\xi$.

Finally, we relate the bounded energy to bounded gradients. The disagreement term in ξ_α is non-negative, so $\alpha \sum_i f_i(x_{(i)}(k)) \leq \xi_\alpha(k)$.

$$\mathbb{E}\left[\sum_{i=1}^{n} f_i(x_{(i)}(k))\right] \leq \frac{1}{\alpha} \mathbb{E}[\xi_\alpha(k)] \leq \frac{B_\xi}{\alpha}$$

Using the property $\|\nabla g(x)\|^2 \leq 2L_g(g(x) - g^*)$ for a convex function g:

$$\begin{aligned} \mathbb{E}[\|h(k)\|^2] = \mathbb{E}\left[\sum_{i=1}^{n} \|\nabla f_i(x_{(i)}(k))\|^2\right] &\leq \mathbb{E}\left[\sum_{i=1}^{n} 2L_h(f_i(x_{(i)}(k)) - f_i^*)\right] \\ &\leq 2L_h \left(\mathbb{E}\left[\sum_{i=1}^{n} f_i(x_{(i)}(k))\right] - \sum_{i=1}^{n} f_i^* \right) \\ &\leq 2L_h \left(\frac{B_\xi}{\alpha} - \sum_{i=1}^{n} f_i^* \right) \end{aligned}$$

This proves that $\mathbb{E}[\|h(k)\|^2]$ is uniformly bounded by a constant $D_q^2 := 2L_h(B_\xi/\alpha - \sum f_i^*)$. □

2.2. **Bounded Deviation from Mean.** With bounded expected gradients, we can now bound the expected deviation of each agent from the network-wide average. Let $\bar{x}(k) = \frac{1}{n}\sum_{i=1}^{n} x_{(i)}(k)$.

Lemma 1 (Bounded Expected Deviation)**.** *Under the assumptions of Theorem 1, the expected squared deviation from the mean is bounded:*

$$\sum_{i=1}^{n} \mathbb{E}[\|x_{(i)}(k) - \bar{x}(k)\|^2] \leq \left(\frac{\alpha D_q}{1-\beta}\right)^2 + \left(\frac{\beta^2 np\Delta^2}{4(1-\beta)^2}\right)$$

under the assumption, $\beta = \max\{|\lambda_2(W)|, |\lambda_n(W)|\} < 1$.

Proof. The global update rule is given by:

$$x_{\text{vec}}(k+1) = (W \otimes I)x_{\text{vec}}(k) - \alpha h(k)$$

Unrolling the global update from $x_{\text{vec}}(0) = 0$ gives:

$$x_{\text{vec}}(k) = -\alpha \sum_{s=0}^{k-1} (W^{k-1-s} \otimes I)h(s) + \sum_{s=0}^{k-1} (W^{k-s} \otimes I)\varepsilon_{\text{vec}}(s)$$

The mean solution vector can be stacked as $\bar{x}_{\text{vec}}(k)$.

$$\bar{x}_{\text{vec}}(k) = \left(\frac{1}{n}\mathbf{1}\mathbf{1}^T \otimes I\right) x_{\text{vec}}(k)$$

$$\implies \bar{x}_{\text{vec}}(k) = -\alpha \sum_{s=0}^{k-1} (\frac{1}{n}\mathbf{1}\mathbf{1}^T W^{k-1-s} \otimes I)h(s) + \sum_{s=0}^{k-1} (\frac{1}{n}\mathbf{1}\mathbf{1}^T W^{k-s} \otimes I)\varepsilon_{\text{vec}}(s)$$

$$\implies \bar{x}_{\text{vec}}(k) = -\alpha \sum_{s=0}^{k-1} (\frac{1}{n}\mathbf{1}\mathbf{1}^T \otimes I)h(s) + \sum_{s=0}^{k-1} (\frac{1}{n}\mathbf{1}\mathbf{1}^T \otimes I)\varepsilon_{\text{vec}}(s)$$

The deviation vector is $(I - J \otimes I)x_{\text{vec}}(k)$, where $J = \frac{1}{n}\mathbf{1}\mathbf{1}^T$.

$$x_{\text{vec}}(k) - \bar{x}_{\text{vec}}(k) = -\alpha \sum_{s=0}^{k-1} ((W^{k-1-s} - J) \otimes I)h(s) + \sum_{s=0}^{k-1} ((W^{k-s} - J) \otimes I)\varepsilon_{\text{vec}}(s)$$

Let the two terms be $A(k)$ and $B(k)$. The cross-term $\mathbb{E}[\langle A(k), B(k)\rangle] = 0$ due to the unbiased nature of the quantization error. Thus, $\mathbb{E}[\|A(k) + B(k)\|^2] = \mathbb{E}[\|A(k)\|^2] + \mathbb{E}[\|B(k)\|^2]$.

$$\begin{aligned}
\mathbb{E}[\|A(k)\|^2] &= \alpha^2 \mathbb{E}\left[\left(\sum_{s=0}^{k-1} \|(W^{k-1-s} - J) \otimes I)h(s)\|\right)^2\right] \leq \alpha^2 \mathbb{E}\left[\left(\sum_{s=0}^{k-1} \|(W^{k-1-s} - J) \otimes I)\|\|h(s)\|\right)^2\right] \\
&= \alpha^2 \mathbb{E}\left[\left(\sum_{s=0}^{k-1} \beta^{k-1-s}\|h(s)\|\right)^2\right] \leq \alpha^2 D_q^2 \left(\sum_{s=0}^{k-1} \beta^{k-1-s}\right)^2 \leq \left(\frac{\alpha D_q}{1-\beta}\right)^2 \\
\mathbb{E}[\|B(k)\|^2] &= \mathbb{E}\left[\left(\sum_{s=0}^{k-1} \|(W^{k-s} - J) \otimes I)\varepsilon_{\text{vec}}(s)\|\right)^2\right] \leq \mathbb{E}\left[\left(\sum_{s=0}^{k-1} \|(W^{k-s} - J) \otimes I)\|\|\varepsilon_{\text{vec}}(s)\|\right)^2\right] \\
&= \mathbb{E}\left[\left(\sum_{s=0}^{k-1} \beta^{k-s}\|\varepsilon_{\text{vec}}(s)\|\right)^2\right] \leq \left(\frac{np\Delta^2}{4}\right)\left(\sum_{s=0}^{k-1} \beta^{k-s}\right)^2 \leq \left(\frac{\beta^2 np\Delta^2}{4(1-\beta)^2}\right)
\end{aligned}$$

Summing these bounds gives the result. □

2.3. **Bounded Gradient Error.** Next, we bound the error between the average of local gradients, $g(k) = \frac{1}{n}\sum_i \nabla f_i(x_{(i)}(k))$, and the gradient at the average point, $\hat{g}(k) = \nabla f(\bar{x}(k)) = \frac{1}{n}\sum_i \nabla f_i(\bar{x}(k))$.

Lemma 2 (Bounded Expected Gradient Error)**.** *Under the same assumptions,*

$$\mathbb{E}[\|g(k) - \hat{g}(k)\|^2] \leq \frac{L_h^2 \alpha^2 D_q^2}{n(1-\beta)^2} + \frac{L_h^2 \beta^2 p \Delta^2}{4(1-\beta)^2}$$

Proof. By Lipschitz continuity and Jensen's inequality:

$$\|g(k) - \hat{g}(k)\|^2 = \left\| \frac{1}{n}\sum_i (\nabla f_i(x_{(i)}(k)) - \nabla f_i(\bar{x}(k))) \right\|^2 \leq \frac{L_h^2}{n}\sum_i \|x_{(i)}(k) - \bar{x}(k)\|^2$$

Taking the expectation and substituting the result from Lemma 1 completes the proof. □

2.4. **Linear Convergence in Expectation.** We now prove the main result: linear convergence of the expected squared error of the mean state to a neighborhood of the optimal solution x^*.

Theorem 2 (Linear Convergence with Quantization)**.** *Let $e_{bar}(k) = \bar{x}(k) - x^*$. Under the assumptions of Theorem 1 and for a sufficiently small $\alpha \leq c_1$, the expected squared error converges linearly:*

$$\mathbb{E}[\|e_{bar}(k+1)\|^2] \leq c_3 \mathbb{E}[\|e_{bar}(k)\|^2] + C_{err}$$

where $c_3 = 1 - 2\alpha c_2 < 1$ and C_{err} is an error floor dependent on α^2, Δ^2, and α^4.

Proof. The mean state evolves as $\bar{x}(k+1) = \bar{x}(k) + \bar{\varepsilon}(k) - \alpha g(k)$, where $\bar{\varepsilon}(k) = \frac{1}{n}\sum_j \varepsilon_{(j)}(k)$. The error evolves as:

$$e_{\text{bar}}(k+1) = e_{\text{bar}}(k) - \alpha\hat{g}(k) + \bar{\varepsilon}(k) - \alpha(g(k) - \hat{g}(k))$$

Taking the norm squared and conditional expectation on $\mathcal{F}_k$, cross-terms involving $\bar{\varepsilon}(k)$ vanish.

$$\mathbb{E}[\|e_{\text{bar}}(k+1)\|^2|\mathcal{F}_k] = \|e_{\text{bar}}(k) - \alpha\hat{g}(k)\|^2 + \mathbb{E}[\|\bar{\varepsilon}(k)\|^2|\mathcal{F}_k] + \alpha^2\|g(k) - \hat{g}(k)\|^2 - 2\alpha\langle \bar{e} - \alpha\hat{g}, g_k - \hat{g}\rangle$$

Using Young's inequality on the cross term -

$$\mathbb{E}[\|e_{\text{bar}}(k+1)\|^2|\mathcal{F}_k] = (1+\alpha\eta)\mathbb{E}[\|e_{\text{bar}}(k) - \alpha\hat{g}(k)\|^2|\mathcal{F}_k] + \mathbb{E}[\|\bar{\varepsilon}(k)\|^2|\mathcal{F}_k] + (\alpha^2 + \frac{\alpha}{\eta})\mathbb{E}[\|g(k) - \hat{g}(k)\|^2|\mathcal{F}_k]$$

The first term is the standard contraction from strong convexity (Lemma 3 from the original paper, which is unchanged):

$$\|e_{\text{bar}}(k) - \alpha\hat{g}(k)\|^2 \leq (1 - \alpha c_2)\|e_{\text{bar}}(k)\|^2$$

The second term is the quantization noise at the mean:

$$\mathbb{E}[\|\bar{\varepsilon}(k)\|^2|\mathcal{F}_k] = \frac{1}{n^2}\sum_j \mathbb{E}[\|\varepsilon_{(j)}(k)\|^2|\mathcal{F}_k] \leq \frac{p\Delta^2}{4n}$$

Taking the full expectation and substituting the bound for $\mathbb{E}[\|g(k) - \hat{g}(k)\|^2]$ from Lemma 3:

$$\begin{aligned}\mathbb{E}[\|e_{\text{bar}}(k+1)\|^2] &\leq (1+\alpha\eta)(1-\alpha c_2)\mathbb{E}[\|e_{\text{bar}}(k)\|^2] + \frac{p\Delta^2}{4n} + \left(\alpha^2 + \frac{\alpha}{\eta}\right)\left(\frac{L_h^2\alpha^2 D_q^2}{n(1-\beta)^2} + \frac{L_h^2\beta^2 p\Delta^2}{4(1-\beta)^2}\right) \\ &= c_3\mathbb{E}[\|e_{\text{bar}}(k)\|^2] + C_{\text{err}}\end{aligned}$$

where $c_3 = (1+2\alpha\eta)(1-2\alpha c_2)$ and C_{err} is the constant collecting all error terms. This recurrence shows that the expected squared error converges geometrically to an error ball of size $C_{\text{err}}/(1-c_3) = C_{\text{err}}/(2\alpha c_2)$. □

Lemma 3 (Bounding B_ξ)**.** *Using the recurrence equation of ξ_α from Theorem 1*

$$\xi_\alpha(k) \leq B_\xi := \xi_\alpha(0) + \frac{\phi}{1-\nu}$$

Proof. Using the energy function recurrence relation from Theorem 1

$$\xi_\alpha(k+1) \leq (1-2c_d\mu_\xi)\xi_\alpha(k) + 2c_d\mu_\xi\xi_\alpha^* + C_q$$

Let $1-2c_d\mu_\xi = \nu$ and $2c_d\mu_\xi\xi_\alpha^* + C_q = \phi$ Unrolling the recurrence we get,

$$\xi_\alpha(k) \leq \nu^k\xi_\alpha(0) + \frac{1-\nu^{k-1}}{1-\nu}\phi$$

Now the upper bound of ξ_α is

$$\sup\big(\nu^k\,\xi_\alpha(0)\big) \;+\; \sup\left(\frac{1-\nu^{k-1}}{1-\nu}\,\phi\right).$$

Putting this in the unrolled relation we get the required bound.

□

2.5. **Local agent convergence.**

Corollary 1 (Local Agent Convergence in Expectation)**.** *Under the assumptions of Theorem 2, the expected error for any individual agent i converges linearly to a neighborhood of the optimal solution x^*. Specifically, there exists a constant C_{local_err} such that:*

$$\mathbb{E}[\|x_{(i)}(k) - x^*\|] \leq c_3^{k/2}\|\bar{x}(0) - x^*\| + C_{local_err}$$

where $c_3 < 1$ is the contraction factor from Theorem 2.

Proof. The proof combines the results for the convergence of the mean (Theorem 2) and the bounded deviation from the mean (Lemma 1).

First, we use the triangle inequality to decompose the total error for agent i:

$$\|x_{(i)}(k) - x^*\| = \|x_{(i)}(k) - \bar{x}(k) + \bar{x}(k) - x^*\| \leq \|x_{(i)}(k) - \bar{x}(k)\| + \|\bar{x}(k) - x^*\|$$

Taking the expectation of both sides, the inequality holds due to linearity:

$$\mathbb{E}[\|x_{(i)}(k) - x^*\|] \leq \mathbb{E}[\|x_{(i)}(k) - \bar{x}(k)\|] + \mathbb{E}[\|\bar{x}(k) - x^*\|]$$

We can relate the expectation of the norm to the expectation of the squared norm using Jensen's inequality. For any random vector X, since the square root function is concave, we have $\mathbb{E}[\|X\|] = \mathbb{E}[\sqrt{\|X\|^2}] \leq \sqrt{\mathbb{E}[\|X\|^2]}$. Applying this to both terms yields:

$$\mathbb{E}[\|x_{(i)}(k) - x^*\|] \leq \sqrt{\mathbb{E}[\|x_{(i)}(k) - \bar{x}(k)\|^2]} + \sqrt{\mathbb{E}[\|\bar{x}(k) - x^*\|^2]}$$

Now we bound each term. The first term is bounded by Lemma 1. The second term's bound comes from unrolling the recurrence in Theorem 2, which shows that $\mathbb{E}[\|\bar{x}(k) - x^*\|^2] \leq c_3^k\|\bar{x}(0) - x^*\|^2 + \frac{C_{\text{err}}}{1-c_3}$.

$$\begin{aligned}\mathbb{E}[\|x_{(i)}(k) - x^*\|] &\leq \sqrt{\left(\frac{\alpha D_q}{1-\beta}\right)^2 + \frac{\beta^2 np\Delta^2}{4(1-\beta)^2}} + \sqrt{c_3^k\|\bar{x}(0) - x^*\|^2 + \frac{C_{\text{err}}}{1-c_3}} \\ &\leq \sqrt{\left(\frac{\alpha D_q}{1-\beta}\right)^2 + \frac{\beta^2 np\Delta^2}{4(1-\beta)^2}} + c_3^{k/2}\|\bar{x}(0) - x^*\| + \sqrt{\frac{C_{\text{err}}}{1-c_3}}\end{aligned}$$

where we used $\sqrt{A+B} \leq \sqrt{A} + \sqrt{B}$. By defining $C_{\text{local_err}}$ as the sum of the constant terms, we obtain the desired result, showing that the expected local error converges linearly to a final error ball. □